Alice P. Curtin

Canadian SKA Scientist at McGill University
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https://curtina.github.io

Research Interests

Fast radio bursts; Pulsars; Radio Telescopes; Magnetars; Galactic Magnetic Field; Very Long Baseline Interferometry; Compact Object Mergers; Interstellar Medium

Research Experience

Canadian SKA Scientist, McGill University — Fall 2025 – Present
Affiliated with the University of Amsterdam/ApI; Advisor: Jason Hessels

CHIME/FRB Operations Co-Lead, McGill University — Summer 2025 – Present
Affiliated with the University of Amsterdam/ApI; Advisor: Jason Hessels

Vanier Canada Graduate Fellow, McGill University — Fall 2022 – Fall 2025
Advisor: Victoria Kaspi as a part of CHIME/FRB

McPhee Fellow / Research Assistant, McGill University — Fall 2019 – Fall 2022
Advisor: Victoria Kaspi as a part of CHIME/FRB

Research Assistant, Carleton College — Jan. 2016 – Aug. 2024
Supervisor: Joel Weisberg with collaboration with Joanna Rankin

REU Research Assistan, University of Utah — June – Aug. 2018
Supervisor: David Kieta as a part of VERITAS & HAWC

Education

Doctor of Philosophy, Physics, McGill University — Sept. 2021 – Aug. 2025
Thesis: *Probing the Origins of FRBs using CHIME: High-energy Counterpart Searches and Burst Morphology*
Advisor: Victoria Kaspi
General GPA: 4.0

Master of Science, Physics, McGill University — Sept. 2019 – Sept. 2021
Thesis: *The Canadian Hydrogen Intensity Mapping Experiment Fast Radio Burst Project: Monitoring the Interference Environment and Studying the Bursting Behaviour of SGR 1935+2154*
Advisor: Victoria Kaspi
General GPA: 4.0

Bachelor of Arts, Physics & Astronomy, Carleton College — Sept. 2015 – June 2019
Thesis: *Jets in Active Galactic Nuclei*
Advisor: Joel Weisberg
General GPA: 3.75

Awards and Recognitions

Canadian SKA Scientist, $600k over 5 years — 2025 – 2030

President's Prize for Public Engagement, as a part of the McGill Eclipse team	2025
Marcel Grossman Award, as part of the CHIME/FRB team	2024
McGill Faculty of Science Funding for Science in Space Outreach Initiative, $5000	2023
Best Graduate Student Talk at CASCA, Penticton, BC:	2023
Honorable Mention for EPO Poster at CASCA, Penticton, BC:	2023
McGill Department of Physics Community Building Award, $1000	2023
Vanier Canada Graduate Fellow, NSERC, McGill University, $150000	2022 – 2025
(Ranked #6 out of all NSERC Vanier candidates in Canada)	
2022 Brockhouse Canada Prize, as part of the CHIME team	2022
Lancelot M. Berkeley – New York Community Trust Prize, as part of the CHIME/FRB team	2022
AAS Berkeley Prize, as part of the CHIME/FRB team	2022
McGill University McPhee Fellowship, $10000	2020
Governor General's Innovation Award, as part of the CHIME team	2020
Distinction in Physics & Astronomy Bachelor Degree, Carleton College	2019
Distinction on Physics Thesis, Carleton College	2019
Honorable Mention for Research on SS 433, University of Utah	2018
Mike Ewers Award, Minnesota Space Grant Consortium, Carleton College, $1000	2018
Minnesota Space Grant Consortium Award, Carleton College, $1000	2016

Publications

Summary: Six first-author refereed publications; four first-author non-refereed contributions; twelve publications as a lead co-author; 40+ total refereed works. Total: 2400 citations; h-index: 22. Last updated Winter 2026.

Refereed First-Author Publications

1. **Curtin, A.P.** on behalf of the SKA Transients SWG (2026), The astrophysics of fast radio bursts, Chapter in Advancing Astrophysics with the SKA – II, *Accepted*

2. **Curtin** et al. (2026), CHIME/FRB Discovery and Localization of the Swift-observed FRB 20241228A, ApJ, 998, 97. doi:10.3847/1538-4357/ae2ea0

3. **Curtin** et al. (2025), Morphology of 20 Repeating Fast Radio Burst Sources at Microsecond Time Scales with CHIME/FRB, ApJ, 992, 206. doi:10.3847/1538-4357/adf844.

4. **Curtin**, Weisberg, & Rankin (2024), Determining the Magnetic Field in the Galactic Plane from New Arecibo Pulsar Faraday Rotation Measurements, ApJ, 975, 215, https://doi.org/10.3847/1538-4357/ad7b15.

5. **Curtin** et al. (2024), Constraining Near-simultaneous Radio Emission from Short Gamma-Ray Bursts Using CHIME/FRB, ApJ, 972, 125, https://doi.org/10.3847/1538-4357/ad5c65.

6. **Curtin** et al. (2023), Limits on Fast Radio Burst-like Counterparts to Gamma-Ray Bursts Using CHIME/FRB, ApJ, 954, 154, https://doi.org/10.3847/1538-4357/ace52f.

Non-Refereed Contributions

1. **Curtin, A.P.** et al. (2026), One Attempt at Building an Inclusive & Accessible Hybrid Astronomy Conference: FRB 2025, doi:10.48550/arXiv.2601.14357

2. **Curtin, A. P.** & Cook A.M. (2025), Fast Radio Bursts 2025, NatAs, 9, 1760. doi:10.1038/s41550-025-02741-1

3. **Curtin, A.P.** and CHIME/FRB Collaboration (2024), "CHIME/FRB Updated Position for Repeating Source FRB 20240316A," ATel 6780, https://www.astronomerstelegram.org/?read=16780.

4. **Curtin, A.P.** and CHIME/FRB Collaboration (2023), "Non-detection of radio emission from GRB 231115A with CHIME/FRB," ATel 16341, https://www.astronomerstelegram.org/?read=16341.

Co-authored Contributions: A full list of my contributions (40+) can be found through ADS. Below, I list only those I was major co-author for.

1. Shin, **Curtin** et al. (2026), CHIME/FRB Discovery of the Extremely Active Fast Radio Burst Source FRB 20240114A, , 997, 334.

2. Anderson et al. **incl. Curtin** (2026), Rapid response triggering for radio transients with the SKA Observatory, Chapter in Advancing Astrophysics with the SKA – II, *Submitted*

3. CHIME/FRB Collaboration incl. **Curtin** (2025), CHIME/FRB Outriggers: Design Overview, ApJ, 993, 55. doi:10.3847/1538-4357/adfdcc.

4. Dong, Kilpatrick, Fong, **Curtin**, et al. (2025), Searching for Historical Extragalactic Optical Transients Associated with Fast Radio Bursts, ApJ, 991, 199. doi:10.3847/1538-4357/adfb74.

5. Dong, Clarke, **Curtin**, et al. (2025), CHIME/Fast Radio Burst/Pulsar Discovery of a Nearby Long-period Radio Transient with a Timing Glitch, ApJL, 990, L49. doi:10.3847/2041-8213/adfa8e.

6. Ng, Pandhi, Mckinven, **Curtin**, et al. (2024), Polarization properties of 28 repeating fast radio burst sources with CHIME/FRB, ApJ, 982, 154. doi:10.3847/1538-4357/adb0bc.

7. Sand, **Curtin** et al. (2024), Morphology of 137 Fast Radio Bursts down to Microseconds Timescales from The First CHIME/FRB Baseband Catalog, accepted in ApJ, https://arxiv.org/abs/2408.13215.

8. Giri, Anderson, Chawla, **Curtin** et al. (2023), Comprehensive Bayesian analysis of FRB-like bursts from SGR 1935+2154 observed by CHIME/FRB, submitted to ApJ, https://arxiv.org/abs/2310.16932.

9. Rankin, Venkataraman, Weisberg, & **Curtin** (2023), Polarization measurements of Arecibo-sky pulsars: Faraday rotations and emission-beam analyses, MNRAS, 524, 5042, https://doi.org/10.1093/mnras/sta

10. CHIME/FRB Collaboration et al. **incl. Curtin** (2021), The First CHIME/FRB Fast Radio Burst Catalog, ApJS, 257, 59, https://doi.org/10.3847/1538-4365/ac33ab.

11. Josephy, Chawla, **Curtin**, et al. (2021), No Evidence for Galactic Latitude Dependence of the Fast Radio Burst Sky Distribution, ApJ, 923, 2, https://doi.org/10.3847/1538-4357/ac33ad.

12. CHIME/FRB Collaboration et al. **incl. Curtin** (2020), A bright millisecond-duration radio burst from a Galactic magnetar, Nature, 587, 54, https://doi.org/10.1038/s41586-020-2863-y.

Proposals

Summary: PI for 302 hours of awarded time with the VLBA, and co-PI on an additional 110 hours with the GBT, VLBA, and VLA.

(Co-PI) *The First Radio Survey of Fast Radio Burst Host Environments* — VLA
PI: Dong & Lazda, Hours Acquired: 60 — 2025

(Co-PI) *Constraining the nebular model of fast radio bursts with three new localizations* — VLBA
PI: Lazda, Hours Acquired: 25 — 2025

(Co-PI) *Unveiling compact radio emission associated with a local universe FRB* — VLBA
PI: Lazda, Hours Acquired: 8 — 2025

(Co-PI) *Confirming Radio Detection of Magnetar SGR J0418+5729* — VLA
PI: Sherman, Hours Acquired: 1.4 — 2025

(Co-PI) *Searching for Radio Emission from the Four 'Low B-Field' Magnetars* — GBT
PI: Sherman, Hours Acquired: 14.25 — 2025

(PI) *Precise Pulsar Positions for CHIME/FRB Outrigger Calibration* — VLBA
PI: Curtin, Hours Acquired: 80 — 2024

(PI) *Precise Pulsar Positions for CHIME/FRB Outrigger Calibration* — VLBA
PI: Curtin, Hours Acquired: 180 — 2023

(PI) *Precise Pulsar Positions for CHIME/FRB Outrigger Calibration* — VLBA
PI: Curtin, Hours Acquired: 42 — 2022

Invited & Contributed Talks

Summary: Eight invited talks; Twelve seminars; 19 contributed or shorter institutional talks; Six posters

Seminars & Colloquia

Cornell University, Astrophysics lunch	Ithaca, NY, 2026
John Abbott College Seminar	Montreal, QC, 2026
DAO Science Tea Seminar	Victoria, BC, 2026
Dominion Radio Astrophysical Observatory Seminar (Invited)	Penticton, BC, 2026
University of Madison-Wisconsin Science Seminar Series	Online, 2026
University of Vermont Physical Society Colloquium (Invited)	Burlington, VT, 2025
Marianopolis College Seminar (Invited)	Montreal, QC, 2024
Institut d'Astrophysique Spatiale Seminar (Invited)	Paris, France, 2024
CIERA, Northwestern University Seminar	Evanston, IL, 2024
Caltech Radio Seminar	Pasadena, CA, 2024
McGill University Seminar (Invited)	Montreal, QC, 2023
Dominion Radio Astrophysical Observatory Tech Talk (Invited)	Online, 2021

Conferences

CASCA 2026, EDI talk	Montreal, QC, 2026
CASCA 2026, Science talk	Montreal, QC, 2026
CASCA 2026, SKA lunch	Montreal, QC, 2026

CASCA 2026, Graduate student day (Invited)	Montreal, QC, 2026
Canadian SKA Community Meeting	Online, 2026
CARTA Meeting (Invited)	Victoria, BC, 2026
Rutgers Transients Conference (Invited)	Online, 2025
Structure and Polarization in the Interstellar Medium	Online, 2024
FRB 2024	Thailand, 2024
CASCA 2024, EDI Talk	Toronto, ON, 2024
FRB 2023	Online, 2023
CASCA 2023	Penticton, BC, 2023
FRB 2022	Busan, SK, 2022
Centre for Research in Astrophysics of Quebec Annual Meeting	Quebec, QC, 2022
RFI 2022, Contribution 1	Online, 2022
RFI 2022, Contribution 2	Online, 2022

Shorter institutional & outreach talks

University of Manchester Transient Journal Club	Manchester, UK, 2026
University of Amsterdam Transient Journal Club	Amsterdam, NE, 2026
McGill University Seminar for High School Students	Montreal, QC, 2025
Rutgers Journal Club	Online, 2025
McGill University Transient Discussion Journal Club	Montreal, QC, 2025
MIT Kavli Institute Tea Talk	Boston, MA, 2024
UC Berkeley Astronomy Lunch Talk	Berkeley, CA, 2024
Astronomy on Tap	Montreal, QC, 2023
Northwestern CIERA Observer's Group Meeting	Online, 2023
WVU Astronomy Journal Club	Online, 2023

Posters

CASCA 2024	Toronto, ON, 2024
FRB 2023	Online, 2023
CASCA 2023	Penticton, BC, 2023
FRB 2022	Busan, SK, 2022
FRB 2021	Online, 2021
American Astronomical Society Annual Meeting	Seattle, WA, 2019

Academic Leadership & Service

Chair of the first Canadian SKA Community Meeting	Fall 2026 – Present
Chair of first iteration of CRASIES[1]	Fall 2026 – Present
LOC for Dynamic Radio Sky	Winter 2026 – Present
LOC for CASCA 2027	Winter 2026 – Present
CHIME/FRB Operations Co-Lead (*Selected position*)	Summer 2025 – Present
Co-convenor of Counterparts Working Group, CHIME/FRB	Winter 2025 – Present

[1] Canadian Radio Astronomy Summer Institute and Extended School

SKA Transient Science Working Group — Winter 2025 – Present
CHIME RFI Committee (*Selected position*) — Fall 2024 – Present
MNRAS, ApJ, SKAO Book, PASA Reviewer (*Selected position*) — Fall 2023 – Present
Co-founder & convener of FRB Early Career Researcher Journal Club — Fall 2023 – Present
NSERC CGRSM Review Committee (*Invited*) — Winter 2026
Co-chair of Local Organizing Committee for FRB 2025 — Fall 2024 – Fall 2025
Co-chair of Scientific Organizing Committee for FRB 2025 — Fall 2024 – Fall 2025
Pipeline Expert & Admin, CHIME/FRB (*Selected position*) — Fall 2023 – Summer 2025
Convener of McGill Transient Discussion — Spring 2022 – Winter 2025

Community Leadership & Engagement

Summary: I have spent over 1500 hours since 2019 on outreach, EDI, and science communication.

CERC EDI Committee, McGill University — Fall 2025 – Present
CASCA Sustainability Committee — Summer 2024 – Present
Co-founder & Principal Member of Science in Space Outreach Initiative, McGill University, Trottier Space Institute, Dell Technologies / Girls Who Game — Spring 2022 – Present
McGill Physics Outreach Committee Member[2] — Spring 2020 – Present
CERC EDI Action Plan Committee, McGill University — Winter – Fall 2025
Co-organizer Graduate Outreach Session for CASCA 2025 (*Invited*) — Spring 2025
Astrobites Admin Committee Member — Fall 2023 – Fall 2024
Astrobites Social Media Chair — Fall 2021 – Fall 2024
Astrobites Climate Change Committee Member — Fall 2021 – Fall 2024
Writer for Astrobites — Winter 2021 – Fall 2024
CIBC Spring Break Camp on Space, Toronto, ON (*Invited*) — 2024, 2025
Judge & Selection Committee Member for CCUWiP Conference (*Invited*) — Winter 2024
Mentorship Panelist for Graduate School (*Invited*), McGill University — Fall 2023
Action Plan Task Force for EDI Committee, McGill Univ. — Summer 2020 – Spring 2021
Multi-National Outreach Alliance, McGill University — Fall 2020
Values Statement Task Force for EDI Committee, McGill University — Summer 2020
Volunteer for Goodsell Observatory, Carleton College — Summer 2016 – Summer 2019
Physics Department Curriculum Committee, Carleton College — Fall 2018 – June 2019
Science Summer Educator, Berkshire Museum, Pittsfield, MA — Summer 2017
Student Leader for Young Summer Astronomy Experience, Carleton College — Summer 2016

Teaching & Mentorship Experience

Summary: Formally mentored four undergraduate students; Informal mentor to 5+ graduate students; 360 hours of teaching experience

Graduate Mentees

Melanie Szpigiell, MSc — Summer 2026 - Present

[2] Outreach initiative coordination and facilitation including public talks, school visits, and large-scale event coordination. I am the longest standing member (out of all faculty, staff, students) on the McGill Physics Outreach Committee. Served as a paid graduate coordinator from 2021–2025.

Sloane Sirota, *External thesis committee member*	Spring 2026 - Present

Undergraduate Mentees

Karina Kaplina, Co-supervised with J. Hessels & A. Cook	Summer 2026
Audrey Bernier, Co-supervised with J. Hessels	Summer 2025, Winter - Summer 2026
Melanie Szpigiell, Co-supervised with V. Kaspi	Summer 2025
Sloane Sirota, Co-supervised with V. Kaspi	Summer 2023 – Winter 2024
Sandhya Rotoo, Co-supervised with V. Kaspi	Summer 2022

Lectures & Organized Courses

Principal Organizer, Canadian Radio Astronomy Summer School (CRASIES)	Summer 2026
Lecture, Canadian Radio Astronomy Summer School (CRASIES)	Summer 2026
Lecture, Radio Astronomy Course, University of British Columbia - Okanagan,	Winter 2026
Co-organizer, Graduate-level AstroStatistics Reading Course, McGill University	Fall 2023
Lecture, Graduate-level AstroStatistics Reading Course , McGill University	Fall 2023

Teaching Assistantships

McGill University

Lab Designer, Facilitator, and Grader for Introductory Physics Course	Summer – Fall 2020
Lab Assistant and Grader for Introductory Electricity and Magnetism	Fall 2019
Lab Assistant and Grader for Introductory Mechanics	Fall 2019

Carleton College

Grader for Math 341, Fourier Series and Boundary Values Problems	Spring 2019
Lab Assistant for Physics 165, Electricity and Magnetism	Winter 2019
Problem Solving Facilitator for First and Second Year Physics	Winter 2019

Skills

Computer Skills: Advanced in Python; Advanced in IDL; Proficient in Unix, Mathematica, and Excel; Experience with C++ and ROOT
Language Skills: Spanish (Proficient); French (Proficient).

Science Communication Articles & Media

1. A. P. Curtin, "Some 'not so fast' fast radio bursts," Astrobites, November 2022.
2. A. P. Curtin, "An FRB way off in the distance," Astrobites, October 2022.
3. A. P. Curtin, "Have we found the origins of fast radio bursts?," Astrobites, September 2022.
4. A. P. Curtin, "Could some short and long gamma-ray bursts have the same parents?," Astrobites, May 2022.
5. A. P. Curtin, "You'll be a limbo star. How (s)low can you go?," Astrobites, February 2022.
6. A. P. Curtin, "Let's get building (some terrestrial planets)!," Astrobites, December 2021.
7. A. P. Curtin, "Another Mysterious Fast Radio Burst Detected… Are We One Step Closer to Discovering their Origins?," Astrobites, November 2021.
8. A. P. Curtin, "New Radio Source Towards the Center of our Galaxy – Say whaaaat," Astrobites, October 2021.

9. A. P. Curtin, “A Fast Radio Burst in a Rather Peculiar Location,” Astrobites, August 2021.
10. A. P. Curtin, “If you had $100 million, how would you look for aliens?,” Astrobites, May 2021.
11. A. P. Curtin, “FRBs are spiraling out of control,” Astrobites, March 2021.
12. A. P. Curtin, “Three Little Outliers in a Sea of Planets, Stars, and Brown Dwarfs,” Astrobites, February 2021.
13. A. P. Curtin, Instagram reel on Nanograv gravitational wave detection, Astrobites, July 2023, 3000 views.
14. A. P. Curtin, Instagram reel on renewable energy in the South Pole, Astrobites, December 2023, 1500 views.
15. A. P. Curtin, Instagram reel on 7 eclipse facts, Astrobites, April 2024, 260,000 views.